MSG No. 196.5 $F231'$ [Type II, cubic]

Table 1: Wyckoff site: `4a`, site symmetry: `23.1'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,7,8,9,10,11,12,49,50,51,52,53,54,55,56,57,58,59,60]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24,61,62,63,64,65,66,67,68,69,70,71,72]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[25,26,27,28,29,30,31,32,33,34,35,36,73,74,75,76,77,78,79,80,81,82,83,84]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[37,38,39,40,41,42,43,44,45,46,47,48,85,86,87,88,89,90,91,92,93,94,95,96]`

Table 2: Wyckoff site: `4b`, site symmetry: `23.1'`

No.	position	mapping
1	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12,49,50,51,52,53,54,55,56,57,58,59,60]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24,61,62,63,64,65,66,67,68,69,70,71,72]`
3	$\left[0,\ \frac{1}{2},\ 0\right]$	`[25,26,27,28,29,30,31,32,33,34,35,36,73,74,75,76,77,78,79,80,81,82,83,84]`
4	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[37,38,39,40,41,42,43,44,45,46,47,48,85,86,87,88,89,90,91,92,93,94,95,96]`

Table 3: Wyckoff site: `4c`, site symmetry: `23.1'`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[1,5,6,14,22,23,27,31,36,40,44,45,49,53,54,62,70,71,75,79,84,88,92,93]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[2,10,11,13,17,18,28,32,33,39,43,48,50,58,59,61,65,66,76,80,81,87,91,96]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[3,7,12,16,20,21,25,29,30,38,46,47,51,55,60,64,68,69,73,77,78,86,94,95]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[4,8,9,15,19,24,26,34,35,37,41,42,52,56,57,63,67,72,74,82,83,85,89,90]`

Table 4: Wyckoff site: `4d`, site symmetry: `23.1'`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[1,5,6,14,22,23,27,31,36,40,44,45,49,53,54,62,70,71,75,79,84,88,92,93]`
2	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[2,10,11,13,17,18,28,32,33,39,43,48,50,58,59,61,65,66,76,80,81,87,91,96]`
3	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[3,7,12,16,20,21,25,29,30,38,46,47,51,55,60,64,68,69,73,77,78,86,94,95]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[4,8,9,15,19,24,26,34,35,37,41,42,52,56,57,63,67,72,74,82,83,85,89,90]`

Table 5: Wyckoff site: `16e`, site symmetry: `.3.1'`

No.	position	mapping
1	$[x,\ x,\ x]$	`[1,5,6,49,53,54]`
2	$[x,\ -x,\ -x]$	`[2,10,11,50,58,59]`
3	$[-x,\ x,\ -x]$	`[3,7,12,51,55,60]`
4	$[-x,\ -x,\ x]$	`[4,8,9,52,56,57]`
5	$\left[x,\ x+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[13,17,18,61,65,66]`
6	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}-x\right]$	`[14,22,23,62,70,71]`
7	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[15,19,24,63,67,72]`
8	$\left[-x,\ \frac{1}{2}-x,\ x+\frac{1}{2}\right]$	`[16,20,21,64,68,69]`

continued ...

Table 5

No.	position	mapping
9	$\left[x+\frac{1}{2},\, x,\, x+\frac{1}{2}\right]$	[25,29,30,73,77,78]
10	$\left[x+\frac{1}{2},\, -x,\, \frac{1}{2}-x\right]$	[26,34,35,74,82,83]
11	$\left[\frac{1}{2}-x,\, x,\, \frac{1}{2}-x\right]$	[27,31,36,75,79,84]
12	$\left[\frac{1}{2}-x,\, -x,\, x+\frac{1}{2}\right]$	[28,32,33,76,80,81]
13	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, x\right]$	[37,41,42,85,89,90]
14	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, -x\right]$	[38,46,47,86,94,95]
15	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, -x\right]$	[39,43,48,87,91,96]
16	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, x\right]$	[40,44,45,88,92,93]

Table 6: Wyckoff site: 24f, site symmetry: 2..1'

No.	position	mapping
1	$[x,\, 0,\, 0]$	[1,2,49,50]
2	$[-x,\, 0,\, 0]$	[3,4,51,52]
3	$[0,\, x,\, 0]$	[5,12,53,60]
4	$[0,\, 0,\, x]$	[6,9,54,57]
5	$[0,\, 0,\, -x]$	[7,11,55,59]
6	$[0,\, -x,\, 0]$	[8,10,56,58]
7	$\left[x,\, \frac{1}{2},\, \frac{1}{2}\right]$	[13,14,61,62]
8	$\left[-x,\, \frac{1}{2},\, \frac{1}{2}\right]$	[15,16,63,64]
9	$\left[0,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[17,24,65,72]
10	$\left[0,\, \frac{1}{2},\, x+\frac{1}{2}\right]$	[18,21,66,69]
11	$\left[0,\, \frac{1}{2},\, \frac{1}{2}-x\right]$	[19,23,67,71]
12	$\left[0,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[20,22,68,70]
13	$\left[x+\frac{1}{2},\, 0,\, \frac{1}{2}\right]$	[25,26,73,74]
14	$\left[\frac{1}{2}-x,\, 0,\, \frac{1}{2}\right]$	[27,28,75,76]
15	$\left[\frac{1}{2},\, x,\, \frac{1}{2}\right]$	[29,36,77,84]
16	$\left[\frac{1}{2},\, 0,\, x+\frac{1}{2}\right]$	[30,33,78,81]
17	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}-x\right]$	[31,35,79,83]
18	$\left[\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	[32,34,80,82]
19	$\left[x+\frac{1}{2},\, \frac{1}{2},\, 0\right]$	[37,38,85,86]
20	$\left[\frac{1}{2}-x,\, \frac{1}{2},\, 0\right]$	[39,40,87,88]
21	$\left[\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$	[41,48,89,96]
22	$\left[\frac{1}{2},\, \frac{1}{2},\, x\right]$	[42,45,90,93]
23	$\left[\frac{1}{2},\, \frac{1}{2},\, -x\right]$	[43,47,91,95]
24	$\left[\frac{1}{2},\, \frac{1}{2}-x,\, 0\right]$	[44,46,92,94]

Table 7: Wyckoff site: 24g, site symmetry: 2..1'

No.	position	mapping
1	$\left[x,\, \frac{1}{4},\, \frac{1}{4}\right]$	[1,14,49,62]
2	$\left[x,\, \frac{3}{4},\, \frac{3}{4}\right]$	[2,13,50,61]
3	$\left[-x,\, \frac{1}{4},\, \frac{3}{4}\right]$	[3,16,51,64]
4	$\left[-x,\, \frac{3}{4},\, \frac{1}{4}\right]$	[4,15,52,63]

continued ...

Table 7

No.	position	mapping
5	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	[5,36,53,84]
6	$\left[\frac{1}{4}, \frac{1}{4}, x\right]$	[6,45,54,93]
7	$\left[\frac{3}{4}, \frac{1}{4}, -x\right]$	[7,47,55,95]
8	$\left[\frac{3}{4}, -x, \frac{1}{4}\right]$	[8,34,56,82]
9	$\left[\frac{3}{4}, \frac{3}{4}, x\right]$	[9,42,57,90]
10	$\left[\frac{1}{4}, -x, \frac{3}{4}\right]$	[10,32,58,80]
11	$\left[\frac{1}{4}, \frac{3}{4}, -x\right]$	[11,43,59,91]
12	$\left[\frac{3}{4}, x, \frac{3}{4}\right]$	[12,29,60,77]
13	$\left[\frac{1}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	[17,48,65,96]
14	$\left[\frac{1}{4}, \frac{3}{4}, x+\frac{1}{2}\right]$	[18,33,66,81]
15	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-x\right]$	[19,35,67,83]
16	$\left[\frac{3}{4}, \frac{1}{2}-x, \frac{3}{4}\right]$	[20,46,68,94]
17	$\left[\frac{3}{4}, \frac{1}{4}, x+\frac{1}{2}\right]$	[21,30,69,78]
18	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	[22,44,70,92]
19	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-x\right]$	[23,31,71,79]
20	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{1}{4}\right]$	[24,41,72,89]
21	$\left[x+\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	[25,38,73,86]
22	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	[26,37,74,85]
23	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	[27,40,75,88]
24	$\left[\frac{1}{2}-x, \frac{3}{4}, \frac{3}{4}\right]$	[28,39,76,87]

Table 8: Wyckoff site: 48h, site symmetry: 11'

No.	position	mapping
1	$[x, y, z]$	[1,49]
2	$[x, -y, -z]$	[2,50]
3	$[-x, y, -z]$	[3,51]
4	$[-x, -y, z]$	[4,52]
5	$[z, x, y]$	[5,53]
6	$[y, z, x]$	[6,54]
7	$[-y, z, -x]$	[7,55]
8	$[-z, -x, y]$	[8,56]
9	$[-y, -z, x]$	[9,57]
10	$[z, -x, -y]$	[10,58]
11	$[y, -z, -x]$	[11,59]
12	$[-z, x, -y]$	[12,60]
13	$\left[x, y+\frac{1}{2}, z+\frac{1}{2}\right]$	[13,61]
14	$\left[x, \frac{1}{2}-y, \frac{1}{2}-z\right]$	[14,62]
15	$\left[-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[15,63]
16	$\left[-x, \frac{1}{2}-y, z+\frac{1}{2}\right]$	[16,64]
17	$\left[z, x+\frac{1}{2}, y+\frac{1}{2}\right]$	[17,65]
18	$\left[y, z+\frac{1}{2}, x+\frac{1}{2}\right]$	[18,66]
19	$\left[-y, z+\frac{1}{2}, \frac{1}{2}-x\right]$	[19,67]
20	$\left[-z, \frac{1}{2}-x, y+\frac{1}{2}\right]$	[20,68]
21	$\left[-y, \frac{1}{2}-z, x+\frac{1}{2}\right]$	[21,69]

continued ...

Table 8

No.	position	mapping
22	$\left[z,\ \frac{1}{2}-x,\ \frac{1}{2}-y\right]$	[22,70]
23	$\left[y,\ \frac{1}{2}-z,\ \frac{1}{2}-x\right]$	[23,71]
24	$\left[-z,\ x+\frac{1}{2},\ \frac{1}{2}-y\right]$	[24,72]
25	$\left[x+\frac{1}{2},\ y,\ z+\frac{1}{2}\right]$	[25,73]
26	$\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}-z\right]$	[26,74]
27	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	[27,75]
28	$\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$	[28,76]
29	$\left[z+\frac{1}{2},\ x,\ y+\frac{1}{2}\right]$	[29,77]
30	$\left[y+\frac{1}{2},\ z,\ x+\frac{1}{2}\right]$	[30,78]
31	$\left[\frac{1}{2}-y,\ z,\ \frac{1}{2}-x\right]$	[31,79]
32	$\left[\frac{1}{2}-z,\ -x,\ y+\frac{1}{2}\right]$	[32,80]
33	$\left[\frac{1}{2}-y,\ -z,\ x+\frac{1}{2}\right]$	[33,81]
34	$\left[z+\frac{1}{2},\ -x,\ \frac{1}{2}-y\right]$	[34,82]
35	$\left[y+\frac{1}{2},\ -z,\ \frac{1}{2}-x\right]$	[35,83]
36	$\left[\frac{1}{2}-z,\ x,\ \frac{1}{2}-y\right]$	[36,84]
37	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	[37,85]
38	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	[38,86]
39	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ -z\right]$	[39,87]
40	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	[40,88]
41	$\left[z+\frac{1}{2},\ x+\frac{1}{2},\ y\right]$	[41,89]
42	$\left[y+\frac{1}{2},\ z+\frac{1}{2},\ x\right]$	[42,90]
43	$\left[\frac{1}{2}-y,\ z+\frac{1}{2},\ -x\right]$	[43,91]
44	$\left[\frac{1}{2}-z,\ \frac{1}{2}-x,\ y\right]$	[44,92]
45	$\left[\frac{1}{2}-y,\ \frac{1}{2}-z,\ x\right]$	[45,93]
46	$\left[z+\frac{1}{2},\ \frac{1}{2}-x,\ -y\right]$	[46,94]
47	$\left[y+\frac{1}{2},\ \frac{1}{2}-z,\ -x\right]$	[47,95]
48	$\left[\frac{1}{2}-z,\ x+\frac{1}{2},\ -y\right]$	[48,96]